

Literature Review

A Hybrid ARIMA XGBoost Model for Forecasting Volatile AI Semiconductor Stocks

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This literature review looks at four key research papers that support our project. Our project builds a two-stage model: first, ARIMA is used to capture the basic trend in stock prices, and then XGBoost is used to fix the errors that ARIMA leaves behind. We apply this to AI semiconductor stocks like NVIDIA, AMD, and Super Micro Computer (SMCI). Each paper below is reviewed across six areas: who wrote it and when, the paper title and where it was published, what the paper does, what makes it stand out, its strengths, and its weaknesses. References: [1, 3, 2, 4].

Table 1: Literature review table of four papers reviewed for the ARIMA XGBoost semiconductor stock forecasting project.

Authors & Year	Paper Title & Where Published	What the Paper Does	What Makes It Special	Strengths	Weaknesses
Adebiyi, A.A., Adewumi, A.O., Ayo, C.K. <i>IEEE Conference, 2014</i> [1]	Stock Price Prediction Using the ARIMA Model <i>16th UKSim-AMSS Int. Conf. on Computer Modelling & Simulation, IEEE</i> DOI: 10.1109/ UK-Sim.2014.67	- The paper tests ARIMA on real stock data from the New York Stock Exchange (Nokia, about 4,000 daily prices) and the Nigeria Stock Exchange (Zenith Bank, about 1,300 daily prices) - It follows a step-by-step process: check if the data is stable using the ADF test, then make it stable by taking differences, then choose the best ARIMA settings using ACF/PACF charts, then fit the model, then pick the best version using BIC score, and finally check that leftover errors look like random noise - Best models found were ARIMA(2,1,0) for Nokia and ARIMA(1,0,1) for Zenith Bank - Results are checked using BIC, R-squared, and error charts	- This is the most complete and clearly explained ARIMA pipeline for stock prediction in IEEE literature. Every step is clearly explained and can be followed by anyone - First paper to use a formal scoring table (BIC) to compare many different ARIMA settings side by side - Shows that ARIMA can predict short-term stock prices just as well as neural networks. This was an important finding at the time - Tests on two different stock markets (US and Nigeria) makes the results more trustworthy	- Published by IEEE so it is trustworthy and peer reviewed - Clear steps that anyone can follow and repeat - BIC scoring stops the model from memorising the data (overfitting) - Tested on two different stock markets - Short-term results are as good as neural networks	- Only uses ARIMA with no second step to fix the leftover errors - ARIMA can only model straight-line patterns, so it misses sudden jumps and clusters of big moves - Data is split into training and test only once with no rolling test over time - Data is from before 2011 so it does not cover AI-era stock behaviour - Does not report RMSE or MAPE, which we use in our project - No tech or chip stocks tested

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Table 1 continued...

Authors & Year	Paper Title & Where Published	What the Paper Does	What Makes It Special	Strengths	Weaknesses
Wang, Yan, Guo, Yuankai <i>China Communications (IEEE Journal), 2020</i> [3]	Forecasting Method of Stock Market Volatility Based on Mixed Model of ARIMA and XG-Boost <i>China Communications, IEEE Journal</i> DOI: 10.23919/JCC.2020.11.016	- The paper builds a three-part model called DWT ARIMA GSXGB - First, it uses a maths tool called DWT (Discrete Wavelet Transform) to split the stock price data into two parts: the smooth overall trend and the noisy error part - ARIMA is then used to predict the smooth trend part; an improved XG-Boost (tuned using grid search) predicts the noisy error part - The two predictions are added back together to get the final stock price forecast - This process is written as: total price = trend part (L_t) + error part (N_t), which matches the formula we use in our own project - Tested on 10 different real stocks, with results compared using MSE and MRE error scores	- This is the first IEEE paper to write the formula $y_t = L_t + N_t$ clearly, with ARIMA handling L_t and XGBoost handling N_t. This is the exact same design as our project - Shows that using XGBoost on ARIMA's leftover errors works better than using XGBoost directly on the raw stock prices - The grid search tuning makes XGBoost noticeably more accurate than just using default settings - Testing on 10 stocks is the most thorough check done for this type of model	- Published in an IEEE journal which is stronger than a conference paper - Proves the combined model beats ARIMA alone, XGBoost alone, and other combinations - Tested on 10 different stocks for reliability - Grid search gives a clear, repeatable way to tune XGBoost - Uses multiple error measures for a full picture	- The DWT step makes the model more complicated than it needs to be. Our project skips this step - Only uses Chinese stock market data so it may not apply to US tech stocks - Does not report MAPE, which we use - The improved XG-Boost (GSXGB) is slightly different from the standard XGBoost we use - No rolling-window testing over time - No chip or AI sector stocks tested
Somkunwar, R.K., Pimpalkar, A., Srivastava, V. <i>IEEE Conference, 2024</i> [2]	A Novel Approach for Accurate Stock Market Forecasting by Integrating ARIMA and XG-Boost <i>IEEE SCEECs 2024</i> DOI: 10.1109/ICAISS58487.2024.10481891	- The paper combines ARIMA and XG-Boost in a simple two-step way with no extra tools like DWT needed - Step 1: ARIMA looks at the past stock prices to find repeating patterns and trends - Step 2: XGBoost is then trained on the errors that ARIMA could not explain, using features like yesterday's price, trading volume, and other market signals from OHLCV data - The paper measures accuracy using RMSE and MAE and shows the combined model does better than ARIMA on its own or XGBoost on its own - Focus is on predicting the direction and trend of stock prices to help with investment decisions	- The most recent (2024) IEEE paper that combines ARIMA and XGBoost exactly like our project does. This shows the idea is still current and valid - The simplest version of the two-step pipeline with no extra pre-processing steps. This is exactly how our model is designed - Gives the strongest recent proof that XGBoost is a better choice than LSTM for fixing ARIMA errors, because it is faster, simpler, and just as accurate - Treats ARIMA and XGBoost as working together rather than competing	- Most recent IEEE paper (2024) which shows the method is up to date - Closest to our exact model design - RMSE and MAE match the error measures we use - Confirms XGBoost beats LSTM at the second step - Simple and clean pipeline that is easy to follow	- Conference paper only so it has less detail than a journal - Does not clearly state which stocks were used - MAPE not reported - No rolling-window test over time - No chip or AI sector stocks - Limited explanation of how XGBoost was tuned

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Table 1 continued...

Authors & Year	Paper Title & Where Published	What the Paper Does	What Makes It Special	Strengths	Weaknesses
Zhang, F., Chen, L., Yu, J. <i>BCP Business & Management</i> , 2022 [4]	A Two-Stage ARIMA Model via Machine Learning and its Application in Stock Price Prediction <i>BCP Business & Management</i> , Vol. 26, pp. 400 to 408 DOI: 10.54691/bcpbm.v26i.1989	- The paper clearly writes the idea as: stock price = straight-line part + left-over part, where ARIMA handles the straight-line part and three different machine learning models (Random Forest, XGBoost, and LSTM) each try to handle the leftover part - Uses a rolling-window approach. This means the model is retrained on a moving block of recent data (100, 150, or 250 trading days) and then tested on the next day, repeated again and again. This is a realistic way to test how the model would work in real life - The leftover errors are scaled (normalised) before being passed to the machine learning models - Tested on three Chinese company stocks over 5 years (about 1,285 daily records per stock) - Performance measured using MSE and MRE	- The two-stage formula where total price equals ARIMA prediction plus error correction is exactly the same structure as our project. This is the paper most similar to what we are building - First paper to fairly compare Random Forest, XGBoost, and LSTM in the second stage, using the same data and the same ARIMA errors as input. This makes the comparison fair - Testing with three different rolling window sizes (100, 150, 250 days) is the most thorough time-based testing in all four papers - The framework is general and flexible so it can be applied to any stock dataset including our NVIDIA/AMD/SMCI data	- Formula matches our project exactly - Compares RF, XGBoost and LSTM fairly in Stage 2 - Rolling-window testing is the most realistic evaluation 5-year, 3-stock test gives broad results - Framework easily applied to our stocks	- Not an IEEE paper as BCP is a smaller journal - Only Chinese stocks tested - Does not report RMSE or MAPE - No chip or AI sector stocks - Does not check if residuals are truly random before passing to ML - Does not address the kind of extreme price swings seen in AI stocks like NVIDIA

What gap does our project fill? Looking at all four papers together, we can see a clear pattern. Paper 1 [1] shows that ARIMA works well as a starting point for stock price prediction. Paper 2 [3] shows that the errors ARIMA leaves behind still contain useful information, and that XGBoost can learn from those errors to improve the overall prediction. Paper 3 [2] shows that a simple ARIMA + XGBoost two-step model (with no extra steps) is still a valid and accurate approach as recently as 2024. Paper 4 [4] shows that testing this type of model with a rolling window, retesting it on new data over time, which gives the most reliable results.

The gap: None of these four papers applies this approach to AI semiconductor stocks such as NVIDIA, AMD, or Super Micro Computer. These stocks are especially volatile because their prices are heavily driven by AI demand, GPU shortages, earnings announcements, and investor hype. All of these cause sudden and large price swings that are harder to predict. Our project is the first to test the ARIMA + XGBoost two-stage model on this specific group of stocks, using daily data from Yahoo Finance and measuring performance with RMSE, MAE, and MAPE.

References

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- [3] Yan Wang and Yuankai Guo. Forecasting method of stock market volatility in time series data based on mixed model of ARIMA and XGBoost. *China Communications*, 17(11):205–221, 2020.
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